Haniyeh Zahra Budaqi

Embedded Software Engineer · Zephyr RTOS · FreeRTOS

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Embedded software engineer writing real-time C/C++ firmware on STM32 and ESP32 with FreeRTOS and Zephyr RTOS. Experienced with CAN, SPI/DMA and BLE interfaces and with designing the PCBs the firmware runs on.

Technical Skills

- **RTOS & Frameworks:** Zephyr RTOS (devicetree overlays, Kconfig, ADC driver API, west), FreeRTOS, ESP-IDF, STM32 HAL / CubeIDE
- **Microcontrollers:** STM32 (F1 series), ESP32 / ESP32-S3, AVR
- **Interfaces & Protocols:** CAN, SPI, I2C, UART, DMA, interrupts, BLE, Bluetooth Classic
- **Programming & Tools:** C, C++, Python, MATLAB/Simulink, Kotlin (Android), Git, CMake
- **Hardware:** Altium Designer (schematic & 2-layer PCB), sensor & data-acquisition integration (load cells, HX711)

Experience

Embedded Software Engineer
Lian Battery, Tehran, Iran — Jan 2026 – Present
- Develop real-time firmware with FreeRTOS on the ESP32-S3 for battery-system applications.
- Designed a modular software architecture for CAN-based communication between subsystems.

MG Adak, Tehran, Iran — Apr 2025 – Jan 2026
- Developed modular STM32 firmware for commercial IoT products, with a focus on reliability and scalability.
- Built an Android (Kotlin) app that configures devices and reads their status over Bluetooth Classic.

Electronics Assistant — Oct 2025 – Jan 2026
Sharif Social Robotics Lab — *Tehran, Iran*
- Designed 2-layer PCBs and implemented BLE communication for real-time telemetry and control of robotic platforms.

Electronics Engineering Intern — Nov 2024 – Jul 2025
Nira System Pouya — *Tehran, Iran*
- Completed specialized training in Embedded C and PCB design, followed by hands-on integration projects.

Selected Projects

IoT Energy Monitoring Node on Zephyr RTOS | **C, Zephyr, ESP32** — Mar 2026 – Apr 2026
- Brought up the Zephyr ADC subsystem on the ESP32 to sample voltage and shunt-current channels and compute real-time power, as part of a team building a LoRaWAN-connected energy-harvesting monitor.
- Configured ADC channels (gain, reference, resolution) in devicetree overlays and accessed them through aliases, so the application code stays board-independent and new boards need only a new overlay.

Non-Blocking RC522 RFID Driver for STM32 | **C, STM32 HAL, SPI + DMA** — 2026
- Wrote an MFRC522 driver in which every register access is an SPI DMA transfer completed by interrupt, so the main loop never stalls.
- Implemented the card-read sequence (REQA, anticollision, BCC check) as a state machine that advances one step per call, with debounced card IDs and a configurable expiry timeout. Code

Head & Power Boards for a Social Robot | **Altium Designer, ESP32** — Oct 2025 – Jan 2026
- Designed two 2-layer PCBs: an ESP32 head board driving ear LEDs and fans, and a power board distributing battery power to the fan, LCD and DC-motor sockets. Both boards worked on the first fabrication run.

Education

B.Sc. Mechanical Engineering — Nov 2020 – Jul 2025
Sharif University of Technology — *Tehran, Iran*
- **GPA:** 3.44/4.00 (16.69/20.00) • **Focus:** Dynamics, Robotics, Control Systems • **Publication:** IEEE ICBME 2025

Certifications & Languages

- CS50 (Harvard) • Spacecraft Dynamics and Control Specialization (CU Boulder) • Intro to MATLAB (Vanderbilt)
- **English:** Fluent (IELTS Academic 8.5/9.0, CEFR C2) • **French:** CEFR A2, actively studying